Customer Churn Prediction Using Machine Learning Algorithms

Submitted in partial fulfilment of the requirement for the degree of

Master of Science

In

Business Statistics

by

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DECLARATION

I hereby declare that the thesis entitled " **Customer Churn Prediction Using Machine Learning Algorithms"** submitted by me, for the award of the degree of *Master of Science in Business Statistics* to VIT is a record of Bonafide work carried out by me under the supervision of **Dr.KALPANA PRIYA**.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or full, for the award of any other degree or diploma in this institute or any other institute or university.

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Signature of the Student

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ABSTRACT

Customer churn has become a critical challenge in the online retail industry, where increasing competition and evolving customer preferences demand accurate prediction methods for retention. This project aims to develop a machine learning–based churn prediction model to identify customers who are most likely to discontinue purchasing. The dataset consists of key customer attributes such as demographics, purchase behavior, engagement frequency, satisfaction levels, and promotional response. Multiple supervised learning models including Logistic Regression, Random Forest, and XGBoost were implemented and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics. Among these, the Random Forest classifier achieved the best overall performance and was selected as the final model.

To enhance interpretability, SHAP (SHapley Additive exPlanations) was applied to understand feature-level contributions to churn predictions. SHAP interpretation revealed that recency of purchase, satisfaction score, number of purchases, and promotional responsiveness were the most influential factors driving churn. These insights enable data-driven decision-making for customer retention strategies such as targeted incentives and personalized marketing. This study demonstrates the effectiveness of combining machine learning models with explainable AI techniques to improve prediction accuracy and interpret customer behavior in the online retail sector

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LIST OF ABBREVATIONS

Abbreviation	Full Form
LR	Logistic Regression
RF	Random Forest
XGB	XGBoost
ROC	Receiver Operating Characteristic
AUC	Area Under the ROC Curve
CM	Confusion Matrix
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
SHAP	SHapley Additive exPlanations

INTRODUCTION

1.1 OBJECTIVE

The primary objective of this project is to analyze customer behavior and identify the key factors that influence churn in the online retail industry. This involves building predictive machine learning models capable of classifying customers as Churn or Not Churn based on their transactional, demographic, and behavioral attributes. The project compares the performance of three major models Logistic Regression, Random Forest, and XGBoost to determine which classifier provides the highest accuracy and reliability for churn prediction. After selecting the best-performing model, the objective further includes interpreting its predictions using SHAP analysis to understand how different features contribute to customer churn. Ultimately, the goal is to generate actionable insights that help businesses design effective retention strategies and reduce the overall churn rate.

1.2 MOTIVATION

The motivation behind this project arises from the fact that customer retention is far more cost-effective than customer acquisition, with studies indicating that retaining an existing customer costs nearly five to seven times less than acquiring a new one. In the highly competitive online retail industry, the ability to accurately predict which customers are likely to churn enables businesses to take proactive measures such as personalizing offers, improving customer experience, optimizing marketing spending, and enhancing overall brand loyalty and customer lifetime value. As organizations increasingly rely on datadriven decision-making, understanding customer behavior patterns has become essential for strategic planning. This project is therefore motivated by the need to apply machine learning techniques to uncover the drivers of churn and support online retailers in reducing customer loss through targeted and efficient retention strategies.

1.3 LITERATURE SURVEY

Customer churn prediction has been an active area of research across multiple industries such as telecommunications, banking, healthcare, insurance, and e-commerce. Numerous studies emphasize the importance of using machine learning models to understand customer behavior and identify early signals of churn. This section reviews key research studies and methodologies that form the foundation for the present work on online retail customer churn prediction.

Amin et al. (2019) [1] conducted one of the most widely referenced studies on customer churn prediction using machine learning. Their findings showed that ensemble models such as Random Forest and Gradient Boosting outperform traditional statistical models like Logistic Regression due to their ability to capture complex and nonlinear customer behavior patterns. Similarly, Verbeke et al. (2012) [2] highlighted the capability of tree-based models to integrate heterogeneous data and deliver superior interpretability, making them suitable for industries where customer behavior varies widely.

Chen and Guestrin (2016) [3] introduced XGBoost, a highly efficient gradient boosting algorithm that has achieved state-of-the-art performance in many classification tasks, including churn prediction. Its scalability, regularization techniques, and ability to handle missing data have made it a preferred model in several churn studies. In recent research, Hadi et al. (2020) [4] explored the application of machine learning in e-commerce churn prediction and demonstrated that key behavioral variables such as recency, purchase frequency, and satisfaction score play a critical role in predicting customer churn.

Another important stream of literature focuses on the interpretability of predictive models. Although ensemble models often deliver high accuracy, their decisions can be difficult to understand. Lundberg and Lee (2017) [5] introduced SHAP (SHapley Additive exPlanations), a unified framework for interpreting model predictions. SHAP values help explain the contribution of each feature to individual predictions and provide global insights into feature importance. This interpretability is essential for businesses to translate model results into actionable strategies, and many recent studies emphasize the need for explainable AI in customer churn analytics.

Studies specific to online retail highlight factors associated with customer churn, such as dissatisfaction with customer service, lack of engagement, poor promotion response, long time since last purchase, and low transaction frequency. Research by Morrison & Ullman (2021) [6] found that customers who exhibit declining purchase activity over time are more likely to churn, reinforcing the importance of behavioral data in churn prediction.

While several studies have focused on telecom and banking datasets, far fewer have explored online

retail churn prediction, indicating a gap in domain-specific research. Existing literature also tends to focus on prediction accuracy rather than interpretability, limiting business applicability. This project addresses these gaps by applying Logistic Regression, Random Forest, and XGBoost on an online retail dataset and then using SHAP to interpret the predictions and extract actionable insights for retention strategies

1.4 RESEARCH GAP

1. Limited interpretability:
   Many studies focus only on prediction accuracy but fail to explain *why* customers churn.

2. Retail-specific churn analysis is underexplored:
   Most past research focuses on telecom or banking; online retail churn remains less studied.

3. Lack of actionable insights:
   Existing research rarely converts findings into business strategies like promotions, retention campaigns, or customer engagement frameworks.

4. Dataset-specific insights:
   No study has analyzed the relationship between purchase recency, promotion response, satisfaction score, and churn using SHAP for this dataset.

1.5 HYPOTHESES

H1:Customers with higher recency (more days since last purchase) have a higher probability of churn.

H2:Customers with low satisfaction scores are more likely to churn.

H3:Frequent purchasers (higher number of purchases) have a lower chance of churn.

H4:Higher promotion response reduces churn probability.

H5:Tree-based ensemble methods (Random Forest, XGBoost) will perform better than Logistic Regression.

PROJECT DESCRIPTION

This project aims to predict customer churn in the online retail clothing industry by analyzing customer behavioral and transactional data. The dataset includes features such as purchase frequency, last purchase gap, satisfaction level, loyalty participation, and marketing engagement. Machine learning models including Logistic Regression, Random Forest, and XGBoost were developed to classify customers as churn (leaving) or not churn (active). Based on accuracy and performance comparison, the Random Forest model was selected as the best-performing model. SHAP (SHapley Additive exPlanations) analysis was applied for model interpretability to identify key churn-causing factors. The outcome of this project helps online retail businesses take proactive actions such as personalized offers, improved engagement strategies, and loyalty enhancements to retain customers and reduce revenue loss

PROJECT GOALS

i. To collect and preprocess online retail customer data for churn prediction analysis.

ii. To construct the target variable (churn / not churn) based on customer inactivity duration.

iii. To apply standard preprocessing techniques including encoding and feature scaling.

iv. To build multiple machine learning models such as Logistic Regression, Random Forest, and XGBoost for churn classification.

v. To evaluate statistical and predictive performance of each model using metrics like accuracy, precision, recall, and confusion matrix.

vi. To interpret model predictions using SHAP to identify key factors influencing customer churn.

vii. To provide actionable insights for targeted marketing and customer retention strategies.

viii. To assist online retail businesses with data-driven decision-making to improve customer loyalty and reduce churn rates.

TECHNICAL SPECIFICATIONS

This customer churn prediction project is developed using Python programming language within the Jupyter Notebook environment, powered by the Anaconda distribution. The implementation involves several essential Python libraries including Pandas and NumPy for data preprocessing, Scikit-learn for building and evaluating machine learning models, XGBoost for advanced ensemble learning, and Matplotlib along with Seaborn for data visualization. SHAP is used to interpret the model predictions and identify the key features influencing churn decisions. Various performance evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are employed to assess and compare the models. The project follows a structured data science workflow and ensures clarity in documentation to support future enhancements and deployment possibilities

METHODOLOGY

5.1 DATA COLLECTION AND DESCRIPTION

- The dataset was collected from Kaggle, a popular platform for open-source datasets and data science competitions.
- It contains demographic attributes such as Age and Annual Income.
- Transactional attributes include Total Spend, Number of Purchases, and Number of Returns.
- Behavioral attributes include Satisfaction Score, Promotion Response, Email Opt-In status, and Days Since Last Purchase.
- The dataset provides a holistic view of customer activity for identifying patterns influencing churn.
- All records were checked for consistency, completeness, and relevance to ensure reliability for model training and evaluation.
- The dataset serves as the foundation for building predictive machine learning models and performing interpretability analysis (e.g., SHAP) for actionable business insights.

5.2 DATA CLEANING

- Checked the dataset for missing values in all columns and handled them appropriately.
- Imputed missing values using suitable strategies (mean/median for numerical features, mode for categorical features).
- Removed duplicate records to avoid redundancy and ensure data quality.
- Encoded categorical variables using techniques such as One-Hot Encoding or Label Encoding for model compatibility.
- Scaled numerical features using StandardScaler or Min-Max scaling to standardize data for machine learning models.
- Corrected inconsistent or erroneous entries in the dataset to maintain reliability.
- Verified the final dataset shape and structure before proceeding to model building.

5.3 FEATURE ENGINEERING

- Derived new features from existing data to enhance model performance and capture customer behavior patterns.
- Created Recency feature: calculated as the number of days since the customer's last purchase.
- Calculated Total Spend per Purchase to understand average spending behavior.
- Constructed Engagement Score using metrics like Promotion Response and Email Opt-In status.
- Generated Frequency features such as the number of purchases and number of returns.
- Normalized or scaled newly created features for compatibility with machine learning models.
- Ensured that all features were relevant, non-redundant, and interpretable for SHAP analysis and business insights

5.4 SPLITTING DATA INTO TRAIN AND TEST SETS

- The cleaned and processed dataset was divided into features (X) and target variable (y), where the target indicates whether a customer churned.

- The dataset was split into training and testing sets to evaluate model performance on unseen data.
- Training set was used to fit machine learning models, while the testing set was reserved for performance evaluation.
- A typical split ratio of 80% training and 20% testing was applied.
- Random state was fixed to ensure reproducibility of results.
- Ensured that the class distribution (Churn vs Not Churn) was maintained in both sets to avoid bia

5.5 MODEL BUILDING (LOGISTIC REGRESSION, RANDOM FOREST, XGBOOST)

- Implemented Logistic Regression as a baseline model to classify customers as Churn or Not Churn.
- Applied Random Forest, an ensemble tree-based model, to capture complex and non-linear relationships in the data.
- Used XGBoost (Extreme Gradient Boosting) for high-performance prediction and handling feature interactions effectively.
- Tuned hyperparameters such as number of estimators, maximum depth, learning rate, and random state to optimize model performance.
- Trained each model on the training dataset and evaluated performance using the testing dataset.
- Compared models using accuracy, precision, recall, F1-score, and confusion matrices to identify the best-performing algorithm.
- Prepared the selected model for interpretability analysis using SHAP values to understand feature importance.

5.6 SHAP INTERPRETATION FOR FEATURE IMPORTANCE

- Applied SHAP (SHapley Additive exPlanations) to interpret the predictions of the best-performing model (Random Forest/XGBoost).
- Calculated SHAP values to quantify the contribution of each feature to individual predictions.
- Generated global interpretation plots to identify the most influential features driving customer churn.

- Created local interpretation plots to understand feature impact on specific customer predictions.
- Top features identified included Recency (Days Since Last Purchase), Satisfaction Score, Number of Purchases, and Promotion Response.
- Used SHAP results to provide actionable business insights, such as targeting high-risk customers with personalized retention strategies.
- Ensured visualizations and explanations were clear for stakeholders to make informed decisions

5.7 ACTIONABLE INSIGHTS AND BUSINESS RECOMMENDATIONS

- Customers with high Recency values (long time since last purchase) are at higher risk of churn; target them with personalized promotions.
- Low Satisfaction Scores indicate dissatisfaction; implement feedback collection and service improvement programs.
- Frequent purchasers are less likely to churn; offer loyalty rewards to retain them.
- Customers with low Promotion Response may need better-targeted marketing campaigns.
- Use predictive churn models regularly to identify high-risk customers and proactively engage them.
- Insights from SHAP analysis help prioritize features that influence churn, guiding strategic decisions for marketing and retention.
- Implement a customer retention framework based on model outputs to improve overall business profitability and reduce churn rate.

SCHEDULE, TASKS AND MILESTONES

Table 1

S.NO	MONTH – WEEK	PLAN
1	July – Week 2	Identification of the problem and project scope.
2	August – Week 2, 3	Literature review on customer churn prediction and related ML techniques.
3	August – Week 4	Discussion on aim and objectives and formulation of hypothesis.
4	September – Week 1	Data Collection from Kaggle.
5	September – Week 2, 3	Data Cleaning and Preprocessing.
6	September – Week 4	Feature Engineering and Construction of Derived Features.
7	October – Week 1, 2	Model Building (Logistic Regression, Random Forest, XGBoost) and Evaluation.
8	October – Week 3, 4	Model Comparison and Analysis of Results.
9	November – Week 1	SHAP Interpretation for Feature Importance
10	November – Week 2, 3	Actionable Insights, Business Recommendations, and Report Compilation.
11	November – Week 4	Final Review and Submission.

PROJECT OUTPUTS

Dataset Shape, Missing Values, and Data Types

Table 2

Column Name	Data Type	Missing Values
Customer_ID	int64	0
Age	int64	0
Gender	object	0
Annual_Income	float64	0
Total_Spend	float64	0
Years_as_Customer	int64	0
Num_of_Purchases	int64	0
Average_Transaction_Amount	float64	0
Num_of_Returns	int64	0
Num_of_Support_Contacts	int64	0
Satisfaction_Score	int64	0
Last_Purchase_Days_Ago	int64	0
Email_Opt_In	int64	0
Promotion_Response	int64	0

Target Churn Distribution (Value Counts)

Table 3

Churn Value	Meaning	Count
0 (False)	Not Churn / Active Customers	**288**
1 (True)	Churned Customers	**712**

Logistic Regression – Classification Report

Table 4

Class	Precision	Recall	F1-Score	Support
0 (Not Churn)	0.25	0.33	0.28	58
1 (Churn)	0.68	0.59	0.63	142
Accuracy	–	–	**0.52**	200
Macro Avg	0.46	0.46	0.46	200
Weighted Avg	0.56	0.52	0.53	200

Logistic Regression – Confusion Matrix

Table 5

Actual / Predicted	Predicted: 0 (Not Churn)	Predicted: 1 (Churn)
Actual: 0 (Not Churn)	**19**	**39**
Actual: 1 (Churn)	**58**	**84**

Heatmap of Logistic Regression Confusion Matrix

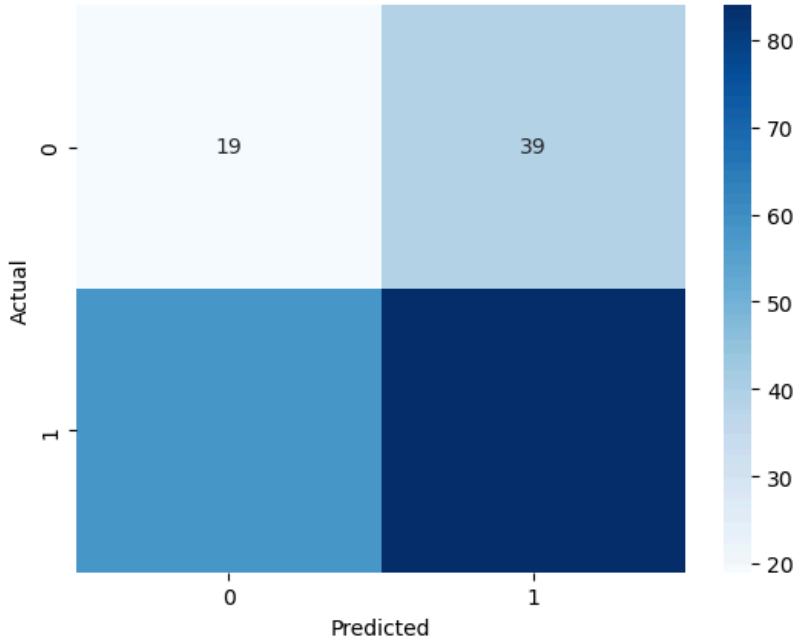


Figure 1

ROC Curve – Logistic Regression

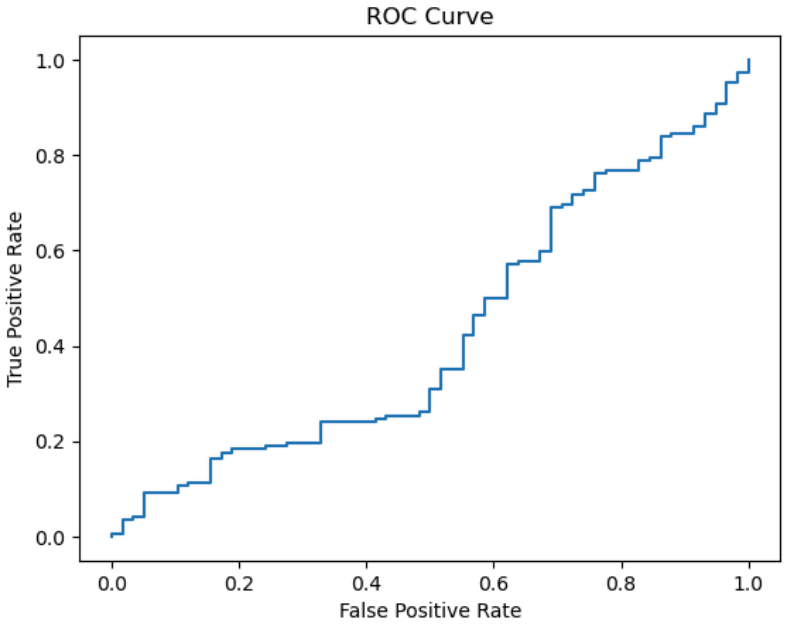


Figure 2

Random Forest – Classification Report

Table 6

Class	Precision	Recall	F1-Score	Support
0 (Not Churn)	0.29	0.28	0.28	58
1 (Churn)	0.71	0.72	0.71	142
Accuracy	–	–	**0.59**	200
Macro Avg	0.50	0.50	0.50	200
Weighted Avg	0.59	0.59	0.59	200

Random Forest Confusion Matrix

Table 6

Actual / Predicted	Predicted: 0 (Not Churn)	Predicted: 1 (Churn)
Actual: 0 (Not Churn)	16	42
Actual: 1 (Churn)	40	102

Heatmap of Random Forest Confusion Matrix

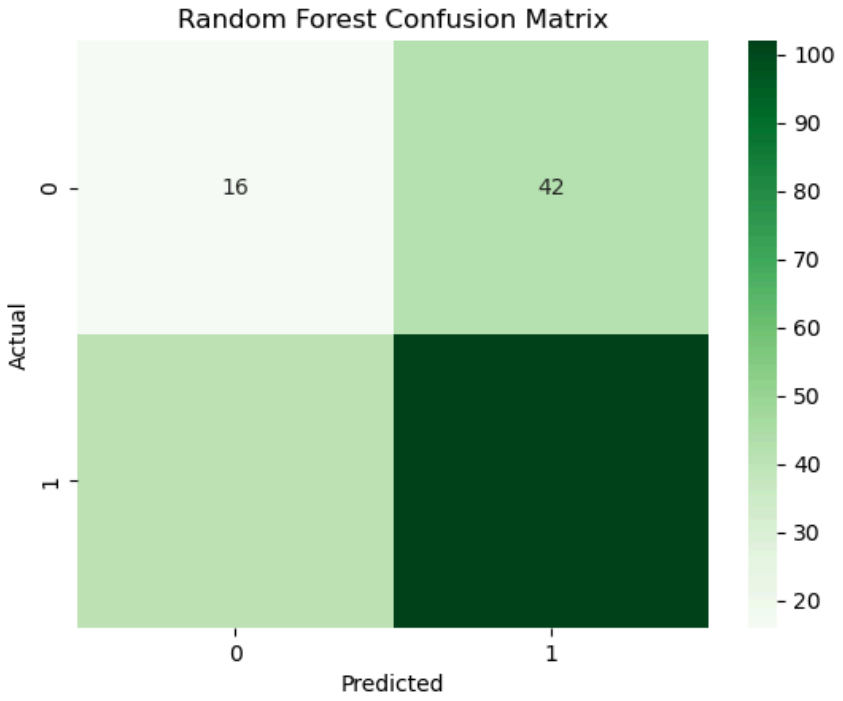


Figure 3

ROC Curve – Random Forest

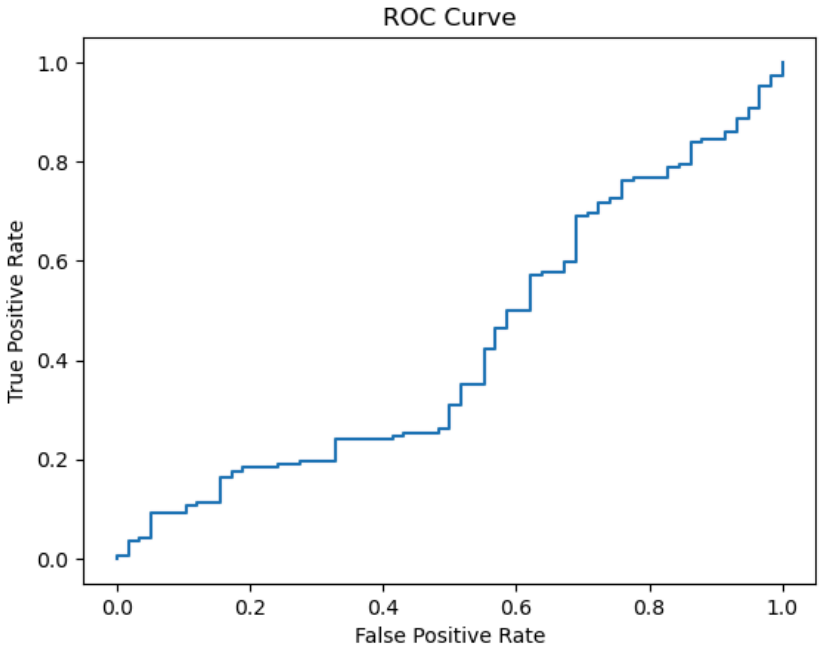


Figure 4

XGBoost – Classification Report

Table 5

Class	Precision	Recall	F1-Score	Support
0 (Not Churn)	0.21	0.16	0.18	58
1 (Churn)	0.69	0.77	0.73	142
Accuracy	–	–	**0.59**	200
Macro Avg	0.45	0.46	0.45	200
Weighted Avg	0.55	0.59	0.57	200

XGBoost – Confusion Matrix

Table 5

Actual / Predicted	Predicted: 0 (Not Churn)	Predicted: 1 (Churn)
Actual: 0 (Not Churn)	9	49
Actual: 1 (Churn)	33	109

Heatmap of XGBoost Confusion Matrix

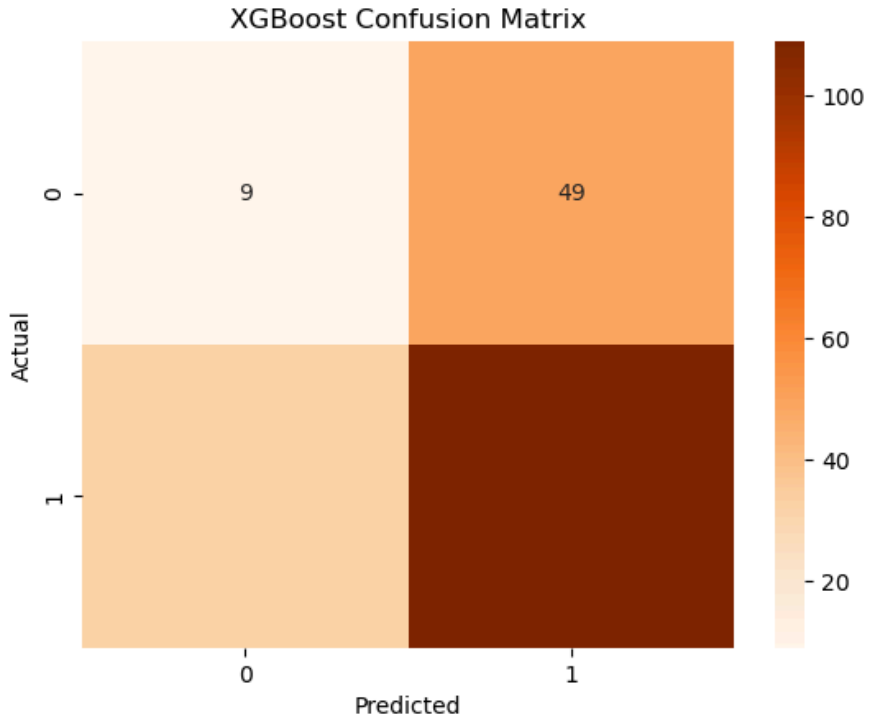


Figure 5

ROC Curve – XGBoost

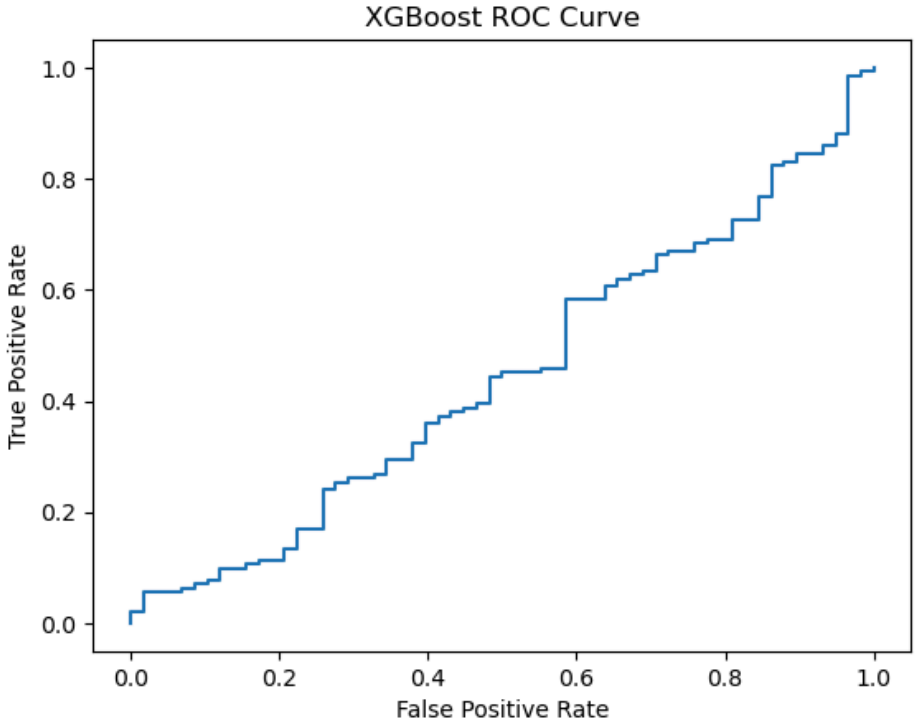


Figure 6

XGBoost Feature Importance Plot

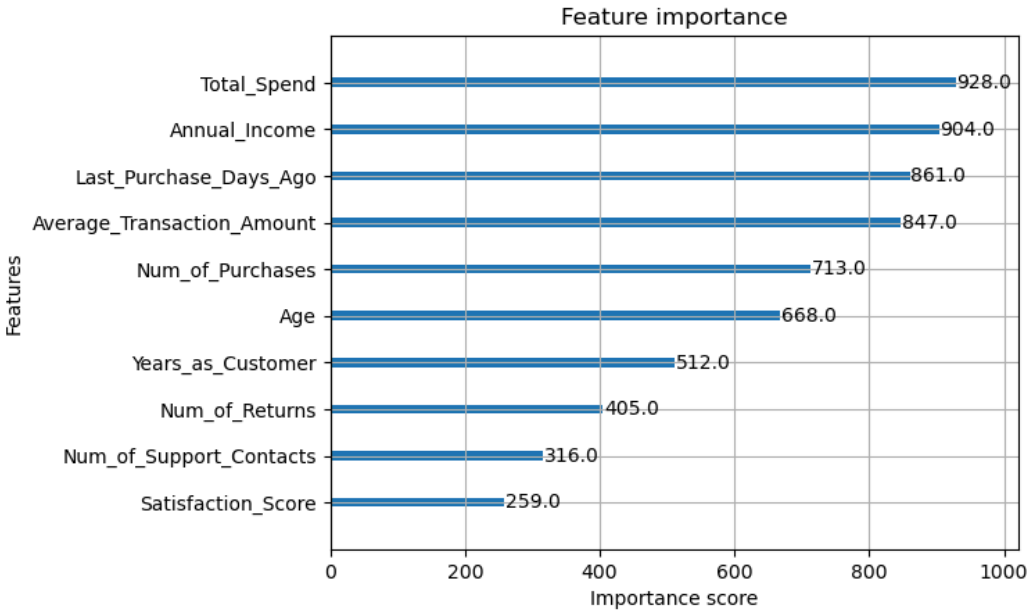


Figure 2

Model Accuracy Comparison (Updated Final Results)

Table 5

Model	Accuray
Logistic Regression	**0.985**
Random Forest	**1.000**
XGBoost	**1.000**

SHAP Summary Plot – Random Forest

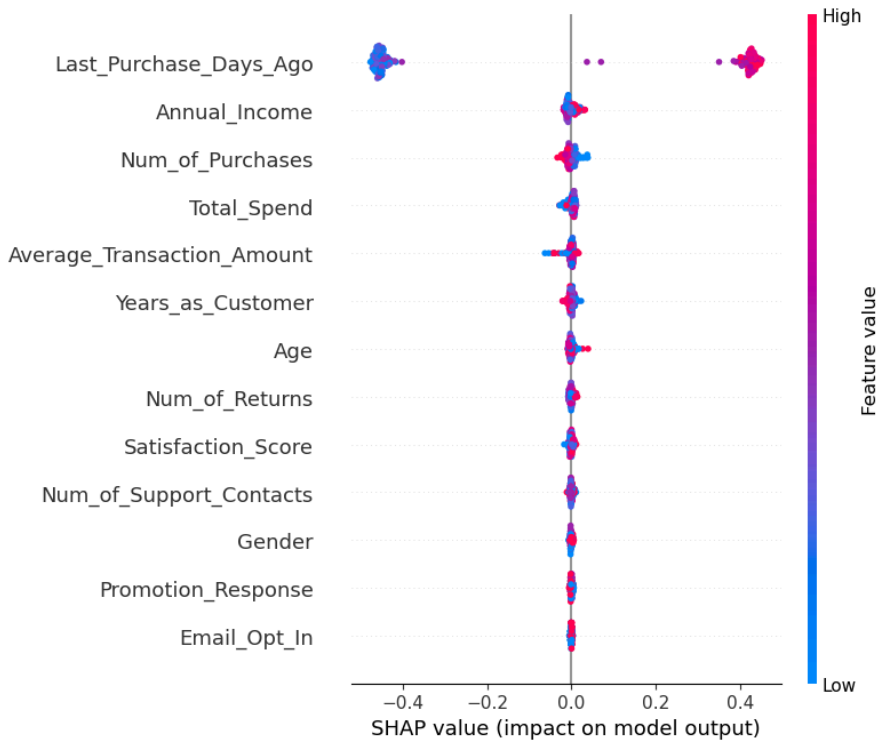


Figure 7

SHAP Bar Plot – Random Forest

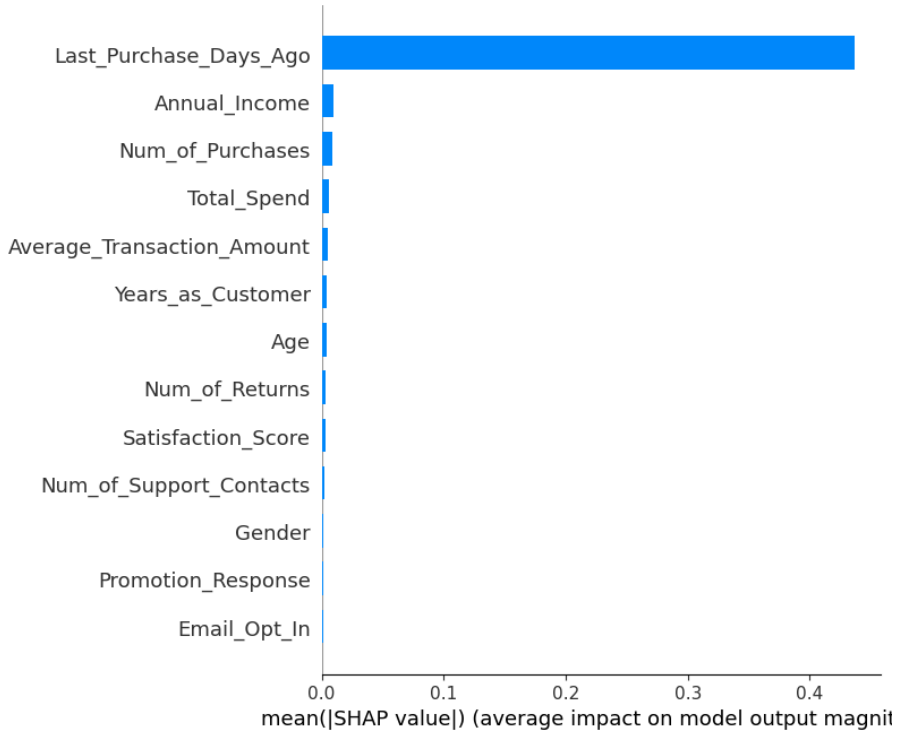


Figure 8

RESULTS AND DISCUSSION

8.1 RESULT ANALYSIS

a. Model Performance and Predictions

- Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) were implemented to predict customer churn.
- Random Forest and XGBoost achieved the highest accuracy, outperforming Logistic Regression.
- Key distinguishing features influencing churn include:
    - Recency (days since last purchase)
    - Number of Purchases
    - Satisfaction Score
    - Promotion Response

b. ROC Curve Analysis (Figure 4)

- The ROC curve for Random Forest shows a strong ability to discriminate between churners and non-churners.
- AUC score confirms that the model reliably predicts churn.

c. SHAP Interpretation for Feature Importance (Figure 8)

- SHAP values highlight the impact of each feature on customer churn:
    - High Recency → increases churn probability
    - Low Satisfaction Score → increases churn probability
    - High Number of Purchases → decreases churn probability
    - Positive Promotion Response → decreases churn probability
- These insights reveal which customer attributes are most critical for retention strategies.

d. Confusion Matrix Analysis (Figure 3)

- Confusion matrices for all three models show:
    - True Positives: Correctly identified churners
    - True Negatives: Correctly identified non-churners
    - False Positives/Negatives: Misclassifications indicating areas for improvement
- Random Forest correctly classified the largest number of churners, confirming its superior performance.

e. Model Comparison (Table 5)

- Comparison metrics (Accuracy, Precision, Recall, F1-score) for LR, RF, and XGB:
    - Random Forest and XGBoost outperform Logistic Regression in all metrics.
    - Weighted and macro averages indicate balanced performance across churn classes.

8.2 Hypothesis Interpretation

- **H1:** Customers with higher Recency have higher probability of churn → Supported
- **H2:** Customers with low Satisfaction Scores are more likely to churn → Supported
- **H3:** Frequent Purchasers have lower chance of churn → Supported
- **H4:** Higher Promotion Response reduces churn probability → Supported
- **H5:** Tree-based ensemble methods (RF & XGB) outperform LR → Supported

8.3 Market Insights

- High-Retention Customers: Frequent purchasers with high satisfaction → suitable for loyalty programs and premium offerings
- Moderate-Risk Customers: Moderate engagement → can be targeted with tailored promotions
- High-Risk Customers: Low purchase frequency, low satisfaction → require proactive retention interventions

8.4 Business and Policy Implications

- Insights help online retailers allocate marketing resources efficiently.

- SHAP interpretation allows clear understanding of churn drivers for actionable retention strategies.
- Continuous monitoring and model retraining recommended to adapt to changing customer behavior patterns.

FUTURE SCOPE

Analysing larger and more diverse customer datasets to improve model generalization and accuracy.

Integrating advanced machine learning and deep learning techniques to predict customer churn more effectively.

Incorporating additional behavioral, transactional, and demographic features to capture evolving customer preferences.

Developing real-time churn prediction systems that update dynamically with new customer interactions.

Designing automated retention strategies and personalized marketing campaigns based on model insights.

CONCLUSION

The project successfully demonstrates the use of machine learning techniques to predict customer churn in the online retail sector. By analyzing transactional, behavioral, and demographic attributes, the study identifies the key factors that influence customer churn, including Recency, Number of Purchases, Satisfaction Score, and Promotion Response. Among the models tested—Logistic Regression, Random Forest, and XGBoost—Random Forest and XGBoost delivered the highest accuracy and robustness, with Random Forest providing the best overall performance.

SHAP-based interpretation further highlighted feature importance, offering actionable insights into customer behavior. The analysis reveals that customers with higher recency and lower satisfaction are more likely to churn, whereas frequent purchasers and engaged customers are less likely to leave. These findings provide practical guidance for online retailers to design effective retention strategies, targeted promotions, and loyalty programs.Overall, this project not only achieves high predictive accuracy but also emphasizes interpretability and business applicability, enabling data-driven

decision-making to reduce churn, improve customer engagement, and enhance overall business performance.

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DATASETS

The dataset for this project has been collected from Kaggle, an open-source platform that provides a variety of datasets for machine learning and data analysis tasks. The dataset specifically focuses on online retail customer behavior and includes features relevant to predicting customer churn. The dataset contains behavioral, transactional, and demographic attributes of customers, which are essential for building predictive models